

Test 1

1

a) Show how to calculate the exact value of $\binom{167}{3}$ **without anything but the basic arithmetic operations** $(+, -, *, /)$ **on a calculator. Show your work.**

b) Write a counting problem for which $\binom{167}{3}$ gives the answer. (Keep your problem simple. Do not use donuts or bit strings in your problem.)

2 Bit strings

a) How many bit strings of length 10 are there?

b) How many bit strings of length 10 have *exactly* 7 1's?

3 A certain state has license plates with 3 letters followed by 3 digits (0 – 9).

a) How many such license plates are there?

b) Terry and Chris are trying to remember a license plate they saw.

- They both remember that all the digits were odd (1, 3, 5, 7, 9).
- Terry remembers that there were two different vowels (A, E, I, O, U) and a consonant (one of the other 21 letters).
- Chris remembers that two of the digits were the same and the other was different.

How many plates match what they remember?

4 Below is an adjacency matrix representation of a graph G .

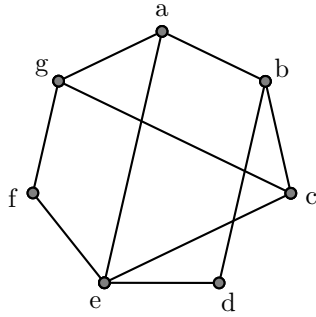
$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

a) How many vertices does G have? How can you tell?

b) Is G directed or undirected? How can you tell?

c) How many edges does G have? How can you tell?

5 Answer the following questions about the graph below.



a) What is an Euler circuit? Does this graph have an Euler circuit? How do you know?

b) Does the graph have an Euler path that is not a circuit? How do you know?

c) What is a bipartite graph? Is this graph bipartite? How do you know?

d) What is a planar graph? Is this graph planar? How do you know?

6 For **one** of K_5 or $K_{3,3}$ (your choice),

a) Draw the graph.

b) Explain how we know that the graph cannot be drawn in the plane with no edge crossings. **DO NOT USE KURATOWSKI'S THEOREM.**

7

a) Show that the two graphs below are isomorphic.

b) **How many isomorphisms** are there between these two graphs? **Explain.**

If you do not know exactly how many there are, tell me as much as you know about how many there are.
Examples: There are at least 3; there are at most 2, etc.

Make sure your answer makes it clear that you know what a graph isomorphism is.

